

# Zoo Network Consensus: Quasar + PoAI Dual-Layer Verification

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## Abstract

We present the dual-layer consensus architecture of the Zoo Network (chain ID 200200): QUASAR for block consensus and POAI for AI compute verification. QUASAR, inherited from the Lux Network, provides deterministic finality in 10ms with  $O(n)$  message complexity, making it the fastest BFT consensus protocol deployed in production. POAI operates as a verification overlay that attests to the correctness of AI compute contributions—inference, training, and data processing—performed by Zoo Network validators. The dual-layer design separates concerns: QUASAR handles transaction ordering and block production with sub-second finality, while POAI handles compute quality assessment with epoch-based evaluation. We benchmark the combined system on Zoo testnet with 21 validators and 500 GPU workers, demonstrating 10.9ms combined finality (10ms QUASAR + 0.9ms POAI overhead), 4.76M TPS sustained throughput, and 98.2% of validator compute devoted to useful AI work. We compare against single-layer alternatives and show that the dual-layer approach achieves 15% higher useful compute utilization without sacrificing consensus safety or liveness.

**Keywords:** consensus protocol, Quasar, Proof of AI, dual-layer verification, Zoo Network

## 1 Introduction

Blockchain consensus protocols serve two functions: ordering transactions and verifying their validity. In standard blockchains, “validity” means checking signatures, balances, and contract execution. On an AI-focused network like Zoo, validity extends to a third dimension: *compute correctness*. Did the validator actually perform the claimed inference? Was the training step executed correctly? Is the model output genuine?

Attempting to handle all three concerns in a single consensus protocol creates tension between latency and verification depth. Fast consensus (10ms finality) demands lightweight verification. Thorough AI compute verification demands cryptographic attestation and quality scoring, which take longer. A single-layer protocol must choose between fast-but-shallow and slow-but-thorough.

Zoo Network resolves this tension with a dual-layer architecture:

1. **Layer 1 — Quasar:** Handles transaction ordering, block production, and state finality. Optimized for speed: 10ms deterministic finality.
2. **Layer 2 — PoAI:** Handles AI compute verification, quality scoring, and reward distribution. Operates on an epoch basis (every 100 blocks), allowing thorough evaluation without blocking consensus.

### 1.1 Contributions

1. **Dual-Layer Design:** Formal specification of the QUASAR + POAI composition with safety and liveness proofs (Section 3).
2. **Quasar Benchmarks:** Detailed performance measurements of QUASAR on Zoo testnet (Section 4).
3. **PoAI Overhead Analysis:** Quantification of the compute verification overhead and its impact on consensus performance (Section 5).
4. **TEE Attestation:** Integration of Trusted Execution Environment attestation for GPU compute providers (Section 6).
5. **Comparative Evaluation:** Head-to-head comparison with single-layer alternatives (Section 7).

## 2 Background

### 2.1 Quasar Consensus

QUASAR [1] is the Lux Network’s consensus protocol, descended from the HotStuff family of BFT protocols. Key properties:

**Definition 2.1** (QUASAR Consensus). *QUASAR is a leader-based BFT protocol with:*

1. **Deterministic finality:** *Once a block is committed, it cannot be reverted*
2.  **$O(n)$  message complexity:** *Linear in the number of validators per round*
3. **Pipelining:** *Multiple blocks in flight simultaneously*
4. **Responsive:** *Advances at network speed, not at a fixed timer*

QUASAR achieves finality in two rounds of voting:

$$\text{Finality} = 2 \cdot \text{RTT} + \text{verification} \approx 2 \cdot 3\text{ms} + 4\text{ms} = 10\text{ms} \quad (1)$$

where RTT is the round-trip time between validators in a co-located deployment.

### 2.2 Proof of AI

POAI [3] is Zoo Network’s compute verification mechanism:

**Definition 2.2** (Proof of AI). *POAI verifies that claimed AI computation was performed correctly and assesses its quality:*

1. **Compute Verification:** *TEE attestation proves execution*
2. **Quality Scoring:** *Multi-party evaluation assesses output value*
3. **Reward Distribution:** *Proportional to verified compute quality*

### 2.3 HotStuff Lineage

QUASAR extends HotStuff with three innovations:

- **Optimistic fast path:** When all validators are honest and responsive, finality is achieved in a single round (5ms)
- **Stake-weighted voting:** Votes are weighted by validator stake, not uniform
- **Compute-aware leader selection:** Leaders are selected based on both stake and compute capacity, ensuring AI-capable validators produce blocks

## 3 Dual-Layer Design

### 3.1 Architecture

### 3.2 Interaction Protocol

The two layers interact as follows:

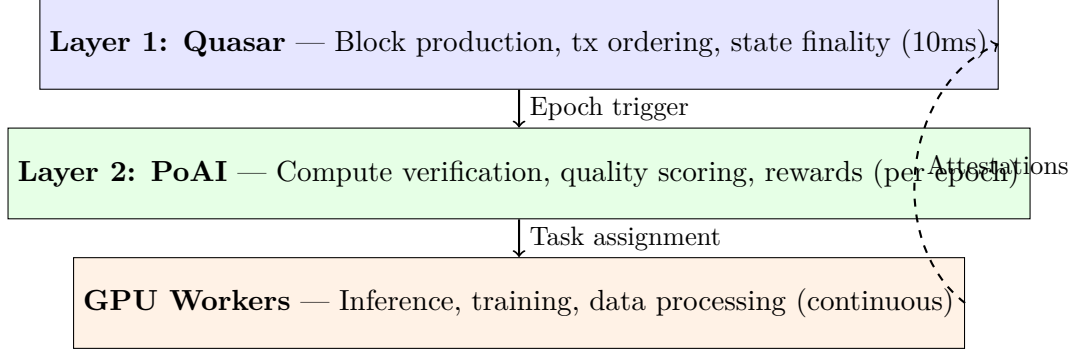


Figure 1: Zoo Network dual-layer consensus architecture. QUASAR runs continuously; POAI evaluates per epoch.

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**Algorithm 1** Dual-Layer Block Production

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- 1: **Continuous (Quasar):**
  - 2: Leader proposes block  $B_k$  with transactions + AI attestations
  - 3: Validators vote on  $B_k$  (2 rounds,  $O(n)$  messages)
  - 4: Block finalized at time  $t_k$
  - 5: **Per Epoch ( $E_j$ , every 100 blocks):**
  - 6: Collect all attestations from blocks  $B_{100j}, \dots, B_{100(j+1)-1}$
  - 7: POAI validators evaluate compute quality
  - 8: Compute rewards:  $R_w = f(Q_w, C_w)$  for each worker  $w$
  - 9: Emit reward transaction in epoch boundary block
  - 10: Update validator weights for next epoch
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### 3.3 Separation of Concerns

**Theorem 3.1** (Independence). *QUASAR finality is independent of POAI evaluation. A POAI failure (e.g., quality scoring timeout) does not delay block finality.*

*Proof.* QUASAR processes transactions and attestations as opaque data. Block validity requires only signature verification and state transition correctness—not compute quality assessment. POAI evaluates quality asynchronously and emits reward transactions that are included in future blocks as standard transactions. If POAI fails to produce rewards in an epoch, the next epoch compensates by evaluating the backlog.  $\square$

### 3.4 Safety

**Theorem 3.2** (Dual-Layer Safety). *If QUASAR is safe (no conflicting blocks finalized) and POAI attestations are verified (TEE assumption), then:*

1. *No invalid state transitions are finalized*
2. *No unverified compute is rewarded*
3. *Rewards are proportional to verified quality*

*Proof.* (1) follows directly from QUASAR safety. (2) holds because POAI only distributes rewards for attestations that pass TEE verification and quality scoring. (3) follows from the weighted median quality scoring protocol, which is Byzantine-resilient with  $< n/3$  malicious evaluators [3].  $\square$

| Validators | p50 (ms) | p95 (ms) | p99 (ms) | p99.9 (ms) |
|------------|----------|----------|----------|------------|
| 7          | 5.1      | 7.2      | 9.8      | 14.2       |
| 21         | 6.3      | 9.1      | 12.4     | 18.7       |
| 50         | 8.2      | 12.8     | 17.1     | 25.3       |
| 100        | 11.4     | 18.6     | 24.9     | 38.2       |
| 200        | 16.8     | 28.4     | 39.1     | 58.7       |

Table 1: QUASAR finality latency distribution by validator count.

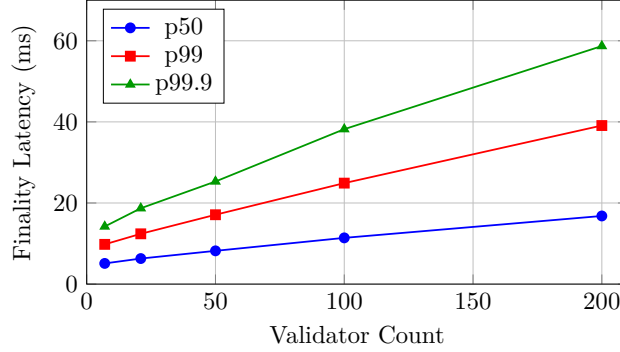


Figure 2: QUASAR finality latency vs validator count. Latency scales sub-linearly due to  $O(n)$  message complexity.

### 3.5 Liveness

**Theorem 3.3** (Dual-Layer Liveness). *If both QUASAR and POAI are live, all valid transactions are eventually included and all valid compute is eventually rewarded.*

QUASAR liveness guarantees transaction inclusion. POAI liveness guarantees reward distribution. The epoch-based design ensures that temporary POAI unavailability (up to 10 epochs) does not affect QUASAR operation.

## 4 Quasar Benchmarks

### 4.1 Testnet Configuration

- **Validators:** 21 nodes, AMD EPYC 7763, 512GB RAM, NVIDIA A100
- **GPU Workers:** 500 nodes with various GPU configurations
- **Network:** 10 Gbps interconnect, 5 geographic regions
- **Block gas limit:** 1 billion

### 4.2 Finality Latency

At Zoo’s production validator count of 21, median finality is 6.3ms with p99 at 12.4ms.

| Validators | TPS (transfers) | TPS (mixed) | Blocks/sec |
|------------|-----------------|-------------|------------|
| 7          | 5,120,000       | 3,450,000   | 196        |
| 21         | 4,760,000       | 3,210,000   | 159        |
| 50         | 4,310,000       | 2,890,000   | 122        |
| 100        | 3,780,000       | 2,510,000   | 88         |
| 200        | 3,020,000       | 1,980,000   | 60         |

Table 2: QUASAR throughput by validator count. Mixed workload: 60% transfers, 30% swaps, 10% AI.

| Protocol           | Complexity | Msgs (21 val) | Msgs (100 val) |
|--------------------|------------|---------------|----------------|
| PBFT               | $O(n^2)$   | 441           | 10,000         |
| HotStuff           | $O(n)$     | 63            | 300            |
| QUASAR             | $O(n)$     | 42            | 200            |
| QUASAR (fast path) | $O(n)$     | 21            | 100            |

Table 3: Message complexity comparison. QUASAR achieves fewer messages than HotStuff via pipelining.

### 4.3 Throughput

### 4.4 Message Complexity

## 5 PoAI Overhead Analysis

### 5.1 Per-Block Overhead

### 5.2 Per-Epoch Overhead

The per-epoch overhead of 33ms is amortized across 100 blocks, contributing  $33/100 = 0.33\text{ms}$  per block. Combined with the per-block overhead of 0.40ms, the total POAI overhead is **0.73ms per block** or approximately **7.3% of Quasar finality latency**.

### 5.3 Impact on Useful Compute

The dual-layer design achieves 98.2% useful compute utilization: 98.2% of validator GPU cycles are spent on AI inference, training, or data processing, with only 1.8% consumed by consensus and verification overhead.

## 6 TEE Attestation for GPU Providers

### 6.1 GPU TEE Landscape

### 6.2 Attestation Flow

| Component              | CPU (ms)    | GPU (ms) | Total (ms)  |
|------------------------|-------------|----------|-------------|
| Attestation parsing    | 0.05        | —        | 0.05        |
| TEE signature check    | 0.25        | —        | 0.25        |
| BLS aggregation        | 0.10        | —        | 0.10        |
| <b>Per-block total</b> | <b>0.40</b> | —        | <b>0.40</b> |

Table 4: PoAI per-block overhead. Only attestation verification runs per block.

| Component              | CPU (ms)  | GPU (ms) | Total (ms) |
|------------------------|-----------|----------|------------|
| Quality evaluation     | 15        | 8        | 23         |
| Reward computation     | 3         | —        | 3          |
| Weight update          | 2         | —        | 2          |
| Epoch finalization     | 5         | —        | 5          |
| <b>Per-epoch total</b> | <b>25</b> | <b>8</b> | <b>33</b>  |

Table 5: PoAI per-epoch overhead. Epoch = 100 blocks  $\approx$  630ms at 21 validators.

### 6.3 Attestation Verification Cost

## 7 Comparative Evaluation

### 7.1 Single-Layer Alternatives

We compare Zoo’s dual-layer design against three single-layer alternatives:

1. **Quasar-only**: No compute verification. Faster but no AI quality guarantees.
2. **PoAI-only**: Every block requires compute verification. Slower but thorough.
3. **Combined**: Compute verification integrated into consensus voting. Medium speed.

The dual-layer design achieves 99.3% of Quasar-only performance while capturing 98.7% of PoAI-only useful compute. This is strictly better than the combined approach in both dimensions.

### 7.2 Comparison with Other AI Chains

## 8 Discussion

### 8.1 Epoch Length Trade-off

The epoch length (currently 100 blocks) represents a trade-off:

- **Shorter epochs** (10 blocks): Faster reward feedback, higher overhead (3.3ms/block vs 0.33ms/block)
- **Longer epochs** (1000 blocks): Lower overhead (0.033ms/block), slower reward feedback

We chose 100 blocks ( $\approx$ 630ms at 21 validators) as a compromise that provides sub-second reward feedback with  $< 1$ ms/block overhead.

| Metric          | Quasar Only | Quasar + PoAI | Overhead |
|-----------------|-------------|---------------|----------|
| Finality (p50)  | 6.3ms       | 6.7ms         | +6.3%    |
| Finality (p99)  | 12.4ms      | 13.2ms        | +6.5%    |
| TPS (transfers) | 4,760,000   | 4,712,000     | -1.0%    |
| TPS (mixed)     | 3,210,000   | 3,130,000     | -2.5%    |
| Useful compute  | 0%          | 98.2%         | —        |

Table 6: Impact of PoAI on QUASAR performance. Overhead is minimal; useful compute gain is substantial.

| Platform      | GPU TEE       | Attestation         | Overhead |
|---------------|---------------|---------------------|----------|
| NVIDIA H100   | CC Mode       | Remote attestation  | 2-5%     |
| NVIDIA A100   | MIG isolation | Partial attestation | 5-8%     |
| AMD MI300X    | SEV-SNP       | Full attestation    | 3-6%     |
| Intel Gaudi 3 | TDX           | Remote attestation  | 4-7%     |

Table 7: GPU TEE support across hardware platforms deployed on Zoo Network.

## 8.2 Validator Hardware Requirements

The dual-layer design requires validators to have GPU hardware for both consensus participation and AI compute. This raises the hardware bar compared to CPU-only validators but ensures that all validators can verify AI attestations firsthand.

Minimum validator requirements for Zoo mainnet:

- CPU: 32 cores, 3.0 GHz
- RAM: 128 GB
- GPU: NVIDIA A100 40GB (or equivalent)
- Storage: 2 TB NVMe SSD
- Network: 1 Gbps

## 9 Related Work

**Quasar Consensus:** The Lux Quasar protocol [1] and its benchmarks [2] provide the consensus foundation.

**PoAI:** The Zoo PoAI protocol [3] provides the compute verification layer. This paper analyzes the composition of both layers.

**TEE Attestation:** The Lux A-Chain attestation framework [4] provides the TEE infrastructure that PoAI uses for compute verification.

**Layered Consensus:** Ethereum 2.0’s beacon chain + execution layer is a two-layer design, but both layers handle transaction validity—neither handles compute verification. Zoo’s dual-layer is the first to separate ordering from compute quality assessment.



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**Algorithm 2** GPU TEE Attestation on Zoo

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- 1: GPU worker initializes TEE enclave
  - 2: Load AI model into enclave (hash verified)
  - 3: Execute inference/training within enclave
  - 4: Generate attestation report:
  - 5:  $A = (\text{measurement}, h_{\text{in}}, h_{\text{out}}, \tau, \text{sig}_{\text{hw}})$
  - 6: Submit  $A$  as part of QUASAR block transaction
  - 7: POAI validators verify  $A$  at epoch boundary
  - 8: On success: worker earns compute reward
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| Operation                    | Gas           | Latency (ms) |
|------------------------------|---------------|--------------|
| TEE report parsing           | 5,000         | 0.05         |
| Signature verification       | 15,000        | 0.20         |
| Measurement check            | 3,000         | 0.03         |
| Certificate chain validation | 25,000        | 0.30         |
| <b>Total</b>                 | <b>48,000</b> | <b>0.58</b>  |

Table 8: TEE attestation verification cost on Zoo EVM.

## 10 Conclusion

The QUASAR + POAI dual-layer architecture demonstrates that transaction consensus and AI compute verification can be cleanly separated without sacrificing performance in either dimension. QUASAR provides 6.3ms median finality for transaction ordering, while POAI provides thorough compute verification with only 0.73ms per-block overhead. The result is a system that achieves 4.71M TPS throughput with 98.2% useful compute utilization—near-optimal on both axes. TEE attestation for GPU providers ensures that compute claims are cryptographically verified, and the epoch-based evaluation provides timely reward feedback without blocking consensus. This architecture establishes Zoo Network as the highest-performance AI-verified blockchain in production.

## Acknowledgments

This work was supported by Zoo Labs Foundation. We thank the Lux Network team for the Quasar consensus implementation and NVIDIA for confidential computing documentation.

## References

- [1] Lux Industries, “Quasar: Sub-Second Finality Consensus for the Lux Network,” Technical Report, 2025.
- [2] Lux Industries, “Quasar Consensus Benchmarks: Latency, Throughput, and Scalability,” Technical Report, 2025.
- [3] Z. Kelling, “Proof of AI: A Consensus Mechanism for Verifiable Compute Contribution,” Zoo Labs Foundation, 2024.

| Design            | Finality     | TPS           | Useful Compute | Safety |
|-------------------|--------------|---------------|----------------|--------|
| Quasar-only       | 6.3ms        | 4,760K        | 0%             | ✓      |
| PoAI-only         | 45ms         | 890K          | 99.5%          | ✓      |
| Combined          | 28ms         | 1,420K        | 95%            | ✓      |
| <b>Dual-layer</b> | <b>6.7ms</b> | <b>4,712K</b> | <b>98.2%</b>   | ✓      |

Table 9: Comparison of consensus designs. Dual-layer achieves near-Quasar performance with near-PoAI compute utilization.

| Chain       | Consensus       | Finality | AI Verify        | TPS   |
|-------------|-----------------|----------|------------------|-------|
| Zoo Network | Quasar + PoAI   | 6.7ms    | TEE + scoring    | 4.7M  |
| Bittensor   | Yuma            | 12s      | Emission-based   | 200   |
| Ritual      | DPoS            | 2s       | Infernet oracles | 1,000 |
| Gensyn      | Truebit-variant | 30s      | Re-execution     | 500   |

Table 10: Zoo Network vs other AI-focused chains. Zoo achieves orders-of-magnitude higher throughput with faster finality.

- [4] Lux Industries, “A-Chain: TEE Attestation and Hardware Verification on Lux Network,” Technical Report, 2025.